

Yesterday we've seen the WSL estimator, with the estimate

$$\hat{x} = \underset{x}{\operatorname{argmin}} \left(\sum_{i=1}^m \frac{1}{\sigma_i^2} \underbrace{(z_i - H_i x)^2}_{r_i = z_i - H_i x} \right)$$

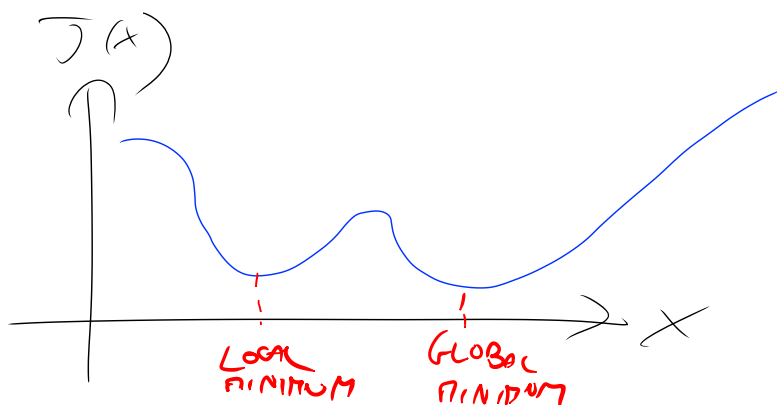
with $z_i - H_i x = z_i - \hat{z}_i$.

Let's now work on the solution.

We have that the optimisation variable is x , so we need to find the best value for x . In an optimisation problem we have, indeed, the target function (the formula inside brackets), as well as some constraints. However, in this specific case, we do not have any constraint. In the end, the objective function is just a function of x , $J(x)$.

What is the idea?

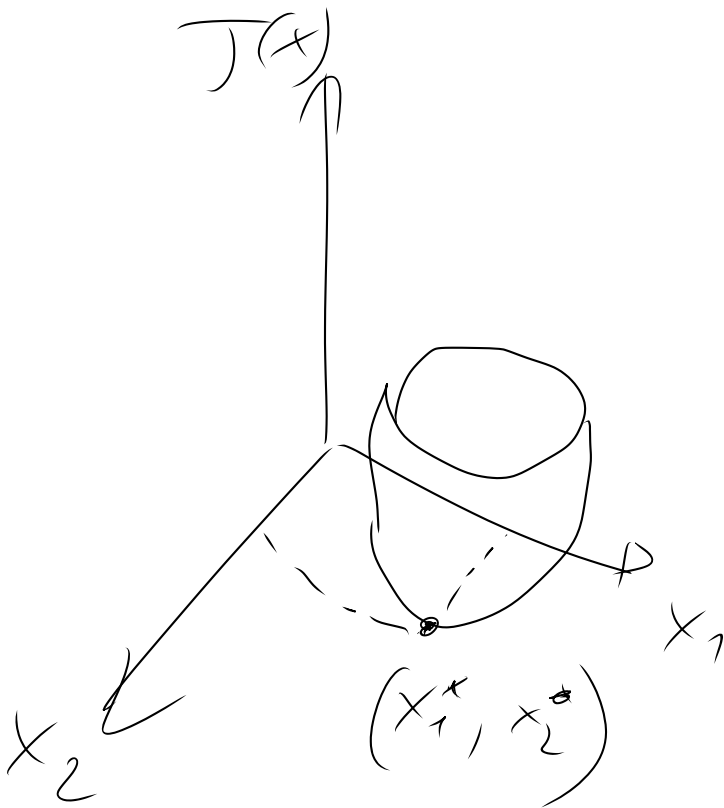
In one dimensional it's quite easy



We want to find the global minimum, so the value such that the function is at it's smaller values. Alongside the global minimum, we also have the local minimum. To find these points, we need to find points such that $\frac{dJ}{dx} = 0$. The points we get will be candidate points. We'll have a convex function, so that it will have just one global minimum.

However, x is a vector of variables. Let's take the case in which we have two variables.

We'll have something like this:



So:

$$\frac{dJ}{dx} = \begin{pmatrix} \frac{dJ}{dx_1} \\ \frac{dJ}{dx_2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So, let's now compute $\frac{dJ}{dx}$, and remember that $J(x) = \sum_{i=1}^m \frac{1}{\sigma_i^2} (z_i - H_i x)^2$. This is the scalar form, if written in matrix form we have:

$$J(x) = (z - Hx)^T \cdot W \cdot (z - Hx) \quad \text{matrix form}$$

with W being the matrix of weights.

$$= (z_1 - H_1 x, z_2 - H_2 x, \dots, z_m - H_m x) \cdot \begin{pmatrix} \frac{1}{\sigma_1^2} & \dots & 0 \\ \vdots & \frac{1}{\sigma_i^2} & \vdots \\ 0 & \dots & \frac{1}{\sigma_i^2} \end{pmatrix} \cdot \begin{pmatrix} z_1 - H_1 x \\ \vdots \\ z_m - H_m x \end{pmatrix}$$

We have that W is symmetric, as it's a diagonal matrix.

$$\begin{aligned} J(x) &= z^T W z + z^T W (-Hx) - (Hx)^T W z - (Hx)^T W (-Hx) = \\ &= z^T W z - z^T W Hx - x^T H^T W z + z^T H^T W Hx = J(x) \end{aligned}$$

The first term does not depend on x , the second and third are linear in x , and the fourth is quadratic. I do one further step, rewriting the third term with a

transpose, as it's just a number. So:

$$\begin{aligned}
 &= z^T W z - 2z^T W H x + x^T H^T W H x = J(x) \\
 \left(\frac{dJ}{dx}\right)^T &= 0 - 2z^T W H + 2x^T H^T W H x \\
 \frac{dJ}{dx} &= -2H^T W z + 2H^T W H x \\
 \frac{dJ}{dx} &= 0 \qquad \qquad \qquad -2H^T W z + 2H^T W H x = 0 \\
 \cancel{2}(H^T W H)x &= \cancel{2}H^T W z
 \end{aligned}$$

We have that H^T is a $n \times m$ matrix, W a $m \times m$ matrix, and H $m \times n$. The product will yield a $n \times n$ matrix.

We multiply both members of the equation by the inverse:

$$\begin{aligned}
 (H^T W H)^{-1} (H^T W H) x &= (H^T W H)^{-1} H^T W z \\
 \hat{x} &= \underbrace{(H^T W H)^{-1} H^T W}_{E \text{ matrix}} \cdot z = E \cdot z
 \end{aligned}$$

So, starting from measurements, we multiply the measurements by matrix E , getting an estimate $\hat{x}$. This E matrix is called the weighted pseudo-inverse of H .

To recap, we have $z = Hx + e$, and with WLS estimation we find

$\hat{x} = (H^T W H)^{-1} H^T W \cdot z = E z$. The estimate is a linear function in the measurements.

Thus, $\hat{z} = H \cdot \hat{x} = \underbrace{H E}_K z = K z$. Also, the estimate measurement is linear in z .

Now, the residual $r = z - \hat{z} = z - K z = (I - K) z$.

Now, practical problem: The estimate $\hat{x}$ could be disrupted by the presence of poor/faulted measurements (bad data).

Typical case: fault of a sensor. How do we deal with this? We need to construct Bad Data Detectors.

We have a vector of measurements z , and maybe z_2 in particular has a problem.

With this measurement we compute the estimate $\hat{x} = \begin{pmatrix} \hat{x}_1 \\ \vdots \\ \hat{x}_n \end{pmatrix}$, and we have as well

$\hat{z} = \begin{pmatrix} \hat{z}_1 \\ \hat{z}_2 \\ \vdots \\ \hat{z}_m \end{pmatrix}$ and $r = \begin{pmatrix} r_1 \\ \vdots \\ r_m \end{pmatrix}$. How do we spot the presence of false data on z_2 ? With

good measurements, the estimated measurement is close to real measurement. If z_2

has a big offset, then $\hat{z}_2$ is very different from z_2 , and thus $r_2 \neq 0$.

Remember that every measurement carry some information about the state, as

$$z_i = H_i x + e_i \quad \forall i = 1, 2, \dots, m.$$

This is a residual based detection, accompanied by a correction procedure, given by the flagging and removing measurements with residuals above a given threshold. So, after the state estimation, we apply this correction procedure, removing measurements with residuals above a certain threshold, like for example $\|r_i\| > \tau$.

After cleaning the data, we get $\hat{x}$, which goes to control modules.

Now, let's discuss the FDIA formulation.

We have measurements z , and somewhere a vector a is injected, effectively corrupting measurements. Both a and z are $\in \mathbb{R}^m$, and sensor i is under attack $\iff a_i \neq 0$.

z_a goes to the SE + BDD module.

The attack equation is simple: $z_a = z + a$.

We now have two questions:

1. Can an attacker inject a particular a , bypassing BDD?
2. If yes, what is the impact on the estimate $\hat{x}_a$?

Normal operation: We have a vector of measurements z with no faults, they go to SE, which produces $\hat{x}$ and r . r goes to BDD, and the BDD has nothing to say as the residual is small.

Now, suppose an attack: We know have z_a sent to the SE. We define $\hat{x}_a$ the estimate computed with corrupted measurements, and r_a the residual. The BDD lets data go through when $\|r_a\| < \tau$. If the attack has no impact on the residual, things will go through (when $\|r\| = \|r_a\|$).

So, for question 1, we need to study (model) the impact of a on r .

If we have $z_a = z + a$, then $\hat{x}_a = E z_a$. Also, $\hat{z}_a = H \hat{x}_a$ and $r_a = (I - K) z_a$.

We focus on $r_a = I z_a - K z_a = (z + a) - K(z + a) = z + a - Kz - Ka$
 $= z - Kz + a - Ka = (I - K)z + (I - K)a$. The first term is exactly the residual with no attack. So, the residual under attack is expressed with the residual, summed to the impact on r , so $r_a = r + (I - K)a$. The problem for the attacker is thus finding a such that $(I - K)a = 0$. This would be a strong/ideal FDIA.

We have that $I - K$ is a known matrix to the operator (but not directly to the attacker). So, $a \in \text{Ker}\{I - K\}$, with $\text{Ker } A = \{x : Ax = 0\}$.

Let's expand $(I - K)a = 0$, getting $(I - HE) = I - \underbrace{H(H^TWH)^{-1}H^TW}_E = I - K$.

We need to find a such that $(I - H(H^TWH)^{-1}H^TW)a = 0$.

If we pick $a = Hc$, we get $Hc - H \cancel{(H^TWH)^{-1}H^TW} Hc = Hc - Hc = 0$.

Condition for strong FDIA is $a = Hc$, for any $c \in \mathbb{R}^n$. If $a = Hc$, $r_a = r$.

So, $z_a = z + a$ will bypass BDD if z does so.

As for question 2, what is the impact of $a = Hc$ on $\hat{x}$?

$$\hat{x} = Ez = (H^TWH)^{-1}H^TWz.$$

We should compute $\hat{x}_a$

$$\begin{aligned}\hat{x}_a &= Ez_a = E(z + a) = Ez + Ea = \hat{x} + Ea \\ Ea &= (H^TWH)^{-1}H^TW \cdot Hc = c \\ \text{so } \hat{x}_a &= \hat{x} + c \text{ (when } a = Hc\text{)}\end{aligned}$$

c is the offset (impact of attack) in state estimation.

So, this models the impact of the attack on the estimate.

FDIA on SE. As for the attack resources, we have:

- Disruption resources: required.
- Disclosure resources: not required
- Model knowledge: required, matrix H .

This was the ideal FDIA. Now let's see generalized FDIA.

We still have an attack on measurements, and this will cause a deviation on the estimates, $\hat{x}_a = \hat{x} + c$, with c being the deviation of estimate caused by attack.

Let's compute the residual $r = z_a - \hat{z}_a = (z + a) - H(\hat{x} + c)$

$$= z + a - H\hat{x} - Hc = z - H\hat{x} + a - Hc = r + a - Hc. \text{ We thus have that}$$

$r_a = r + a - Hc$. With strong FDIA, $a = Hc$ and $r_a = r$. What is important for detection is $\|r_a\|$. We have that $\|r_a\| = \|r + (a - Hc)\| \leq \|r\| + \|a - Hc\|$ by triangle inequality, and also $\leq \tau$, with it being the determinant threshold. Any attack bypassing $\|a - Hc\| \leq \tau - \|r\|$ will bypass the BDD.

Now, let's study a bit the practical construction of the attack. In practice, an attacker will be able to attack only some sensors. We define the attack set $K \subseteq \{1, 2, \dots, n\}$ as the subset of sensors the attacker can corrupt.

We have $a = Hc$, with a the vector of False Data injected into measurements. We also have that $a_i = 0 \forall i \notin K$, so all the meters not belonging to the attack set must be equal to zero. No false data can be injected in the meters that are not under control of the attacker.

But before, let's see an intuition on the undetectability condition. In normal operation, we have $z = Hc$. The SE will tell us $\hat{x} = x$. If we have an attack, we have $z = Hx + Hc = H(x + c)$. This values go to SE, but this cannot be distinguished from a normal operation where the state is $x + c$.

Now, given a system, how can we quantify its vulnerability to this attack? We can introduce the idea of security indices.

We discuss first the sparsity of an attack a . It is the number of components of attack vector a different from zero. This is, in practice, the number of attacked meters.

The sparsity is defined by the zero-norm of vector a ($\|a\|_0$), which is the number of elements of a different from zero. So, to quantify the vulnerability of the network, we need to compute the minimum sparsity attack.

The first indicator is K^* indicator, this is the minimum number of meters necessary to launch strong FDIA. We have that $K^* = \min_c \|a\|_0$, subject to $a = Hc$.

There will be a particular choice of c that will result in the smallest possible $\|a\|_0$, which is the minimum number of meters to attack.

If K^* is small, the system is very vulnerable.

Let's see two local vulnerability indicators.

α_i - we can compute it for every meter $i = 1, 2, \dots, n$

Question: assume I (attacker) can control meter i , how many more meters do I need to control to launch strong FDIA? Again, this is a question on minimum sparsity attack, working on meter i .

We have that $\alpha_i = \min \|a\|_0$ subject to $a = Hc$ (undetectability condition), with $c \neq 0$ (non-triviality) and $a_i \neq 0$.